Protein Engineering-
how can you build wonderful new proteins?





Mark Howarth,
Department of Pharmacology

Overview

- This lecture considers:
- Why engineer proteins?

  stability/folding, expression, *activity*
- Harnessing natural diversity
- Rational design
- Directed evolution
- De novo design

- By the end of this lecture you should know:
- How to change protein function by truncating or fusing
- Sensitivity of proteins to size and post-translational modification
- Different ways to generate antibodies from animals or in the test-tube

What to engineer about a protein?

Size

Folding/stability

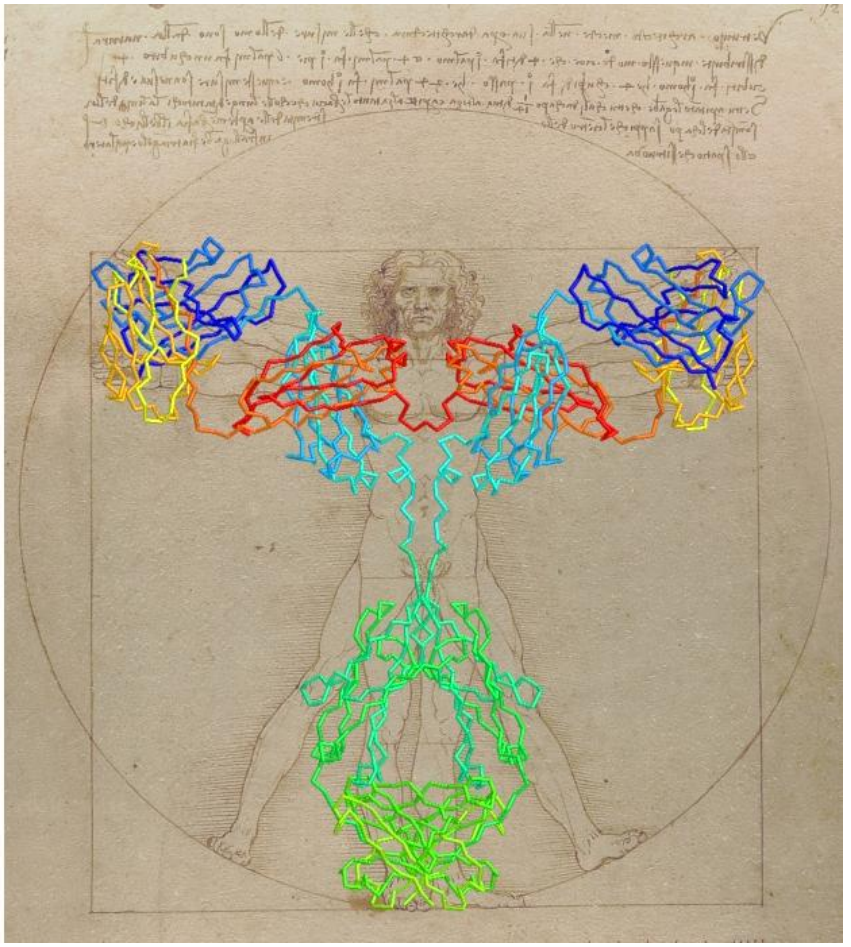
Fusions

Immunogenicity

Activity

Post-translational modification

Switchable (“smart”)



Why are antibodies so important in protein engineering?

Dominant class of new therapeutics
- stable
- well-expressed
- multi-functional
- $$$

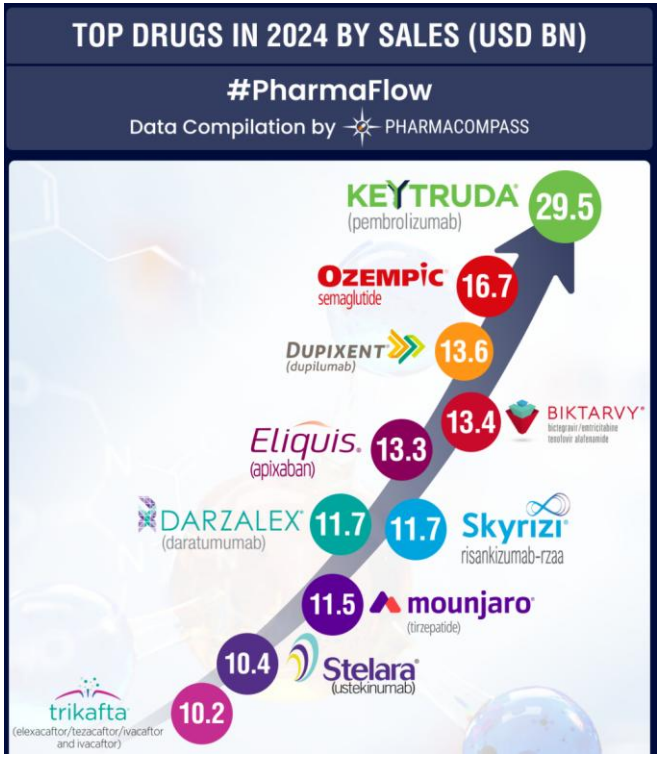
Diagnostics

Foundation of vaccine success

Basic research

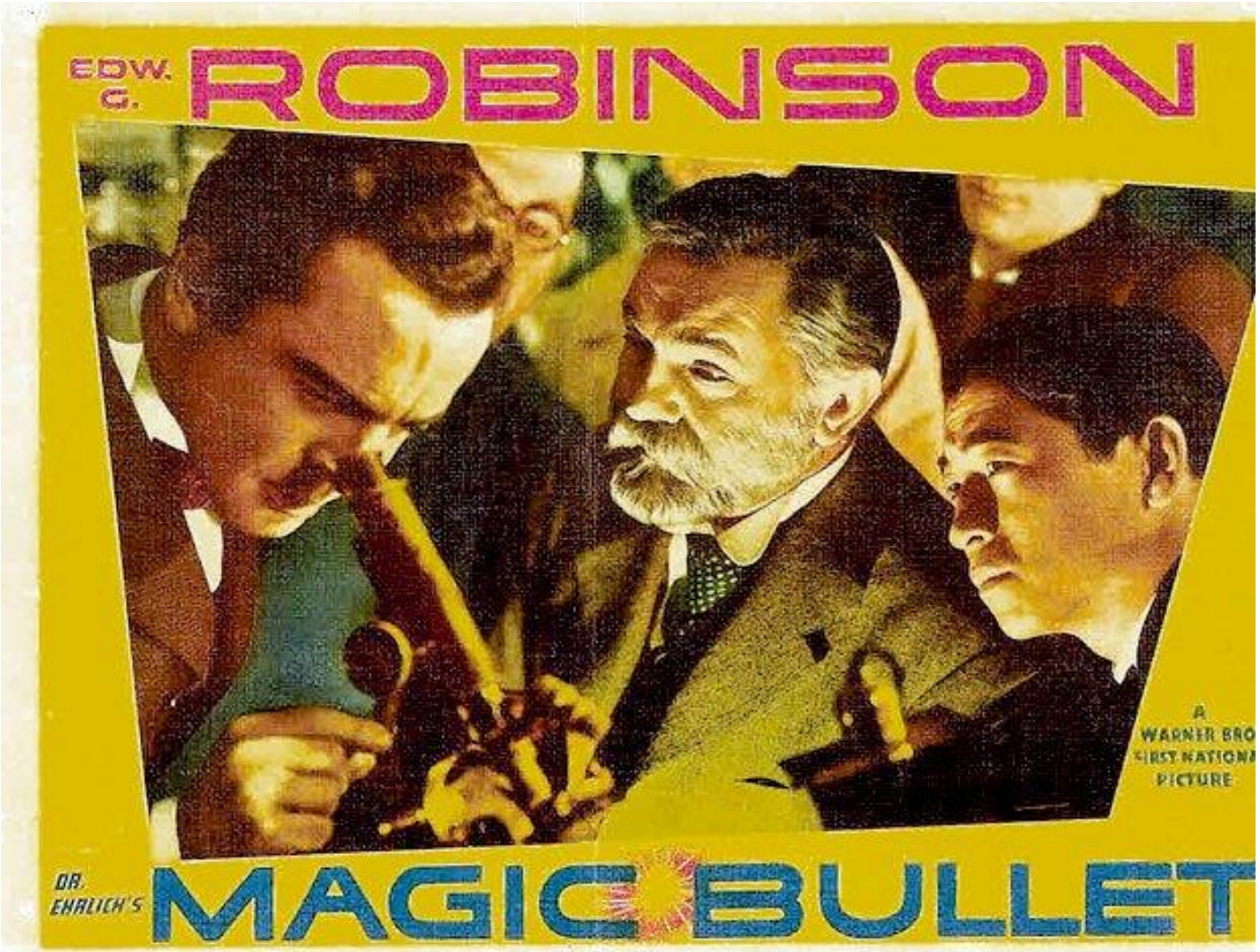




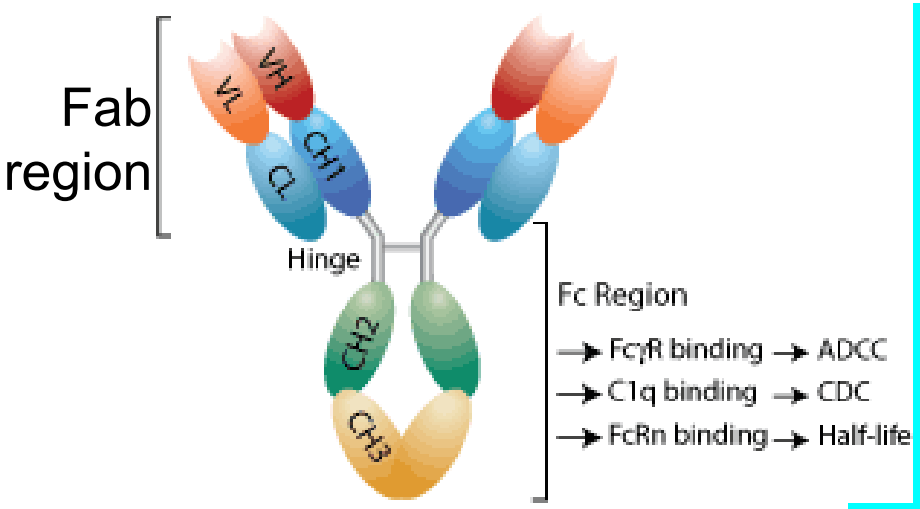


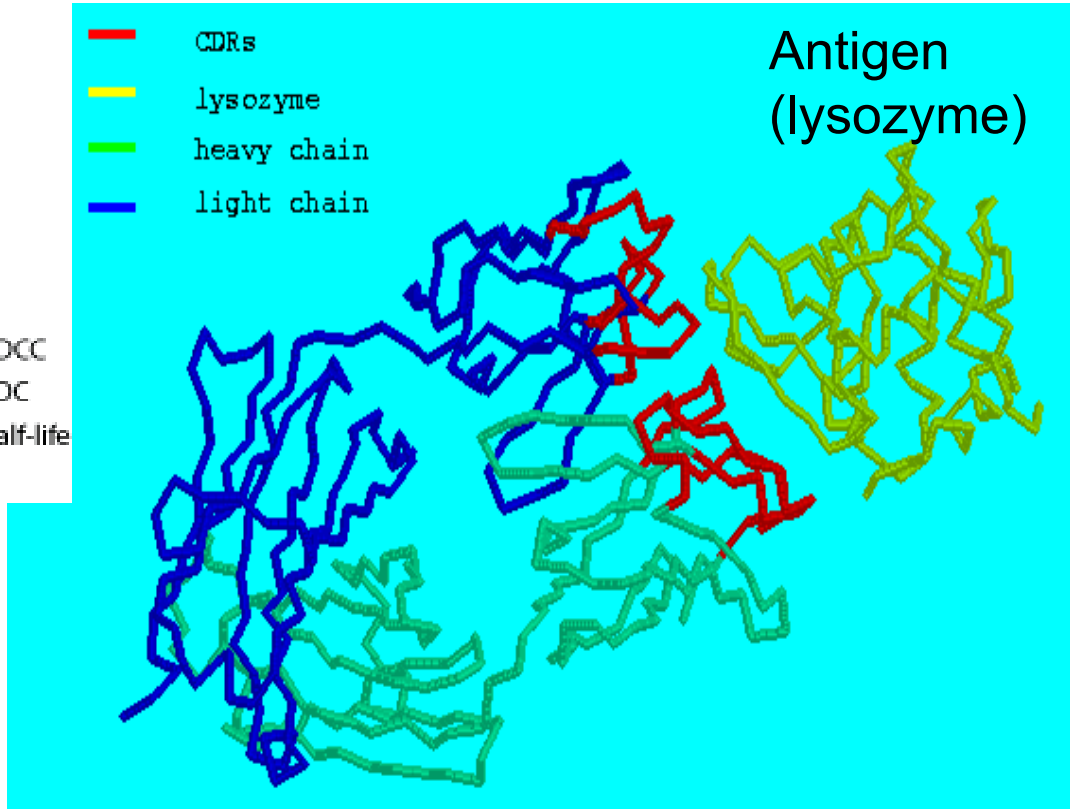
Antibody therapy

Dreaming of the Magic Bullet



Antibody structure

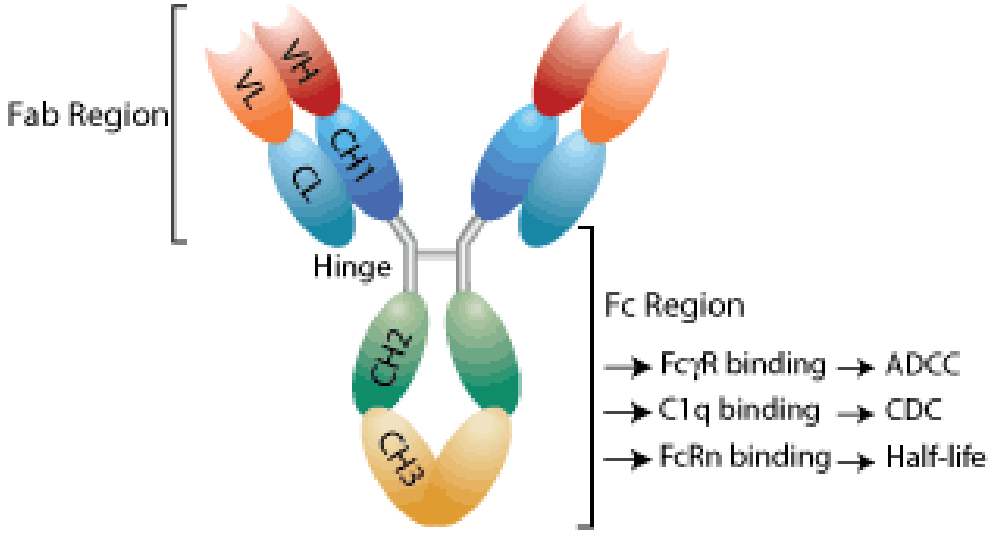


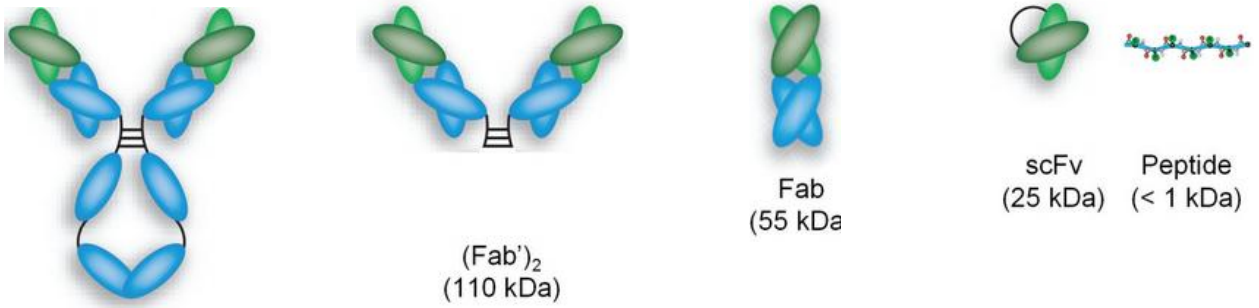


Variation limited to 3 loops at tips of heavy chain and light chain: Complementarity Determining Regions (CDRs) for target binding.

What to engineer: Size

Domain structure of IgG antibody





Ab fragments with different size:
clearance from blood
penetration into tumours
resistance to misfolding/proteolysis

Why do CDRs expressed as peptides not bind tightly to targets?

(A) Proteolytically sensitive

(B) Non-specific

(C) Too flexible

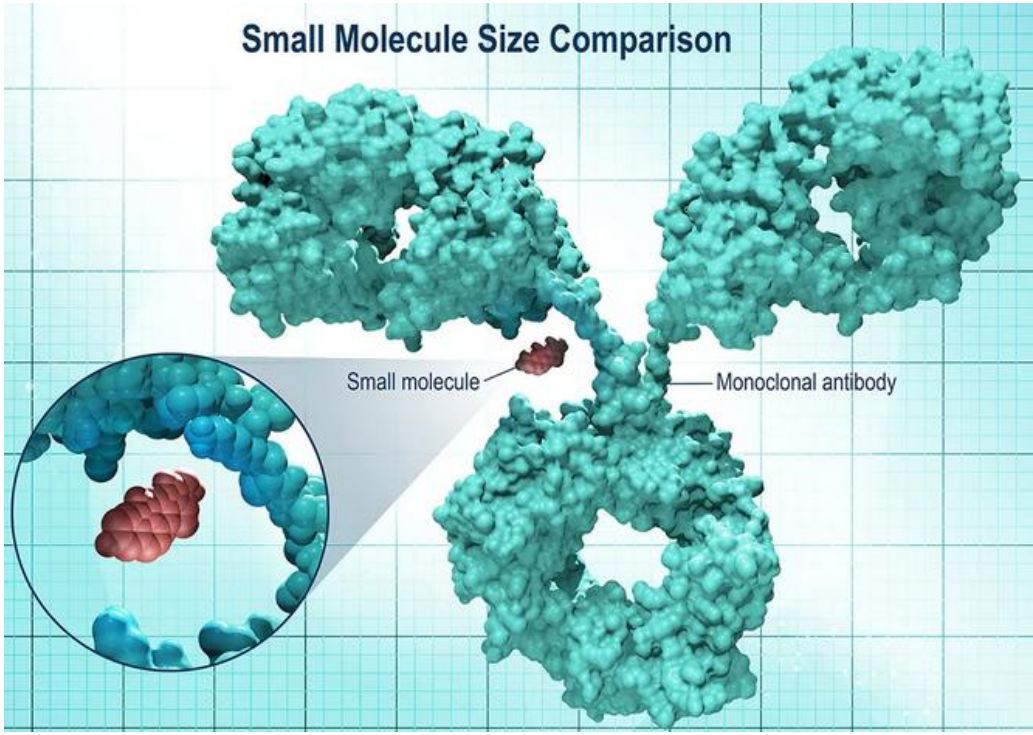
(D) Wrong charge

Antibody versus small molecule therapeutics

Why use antibodies when you can use small molecule drug?

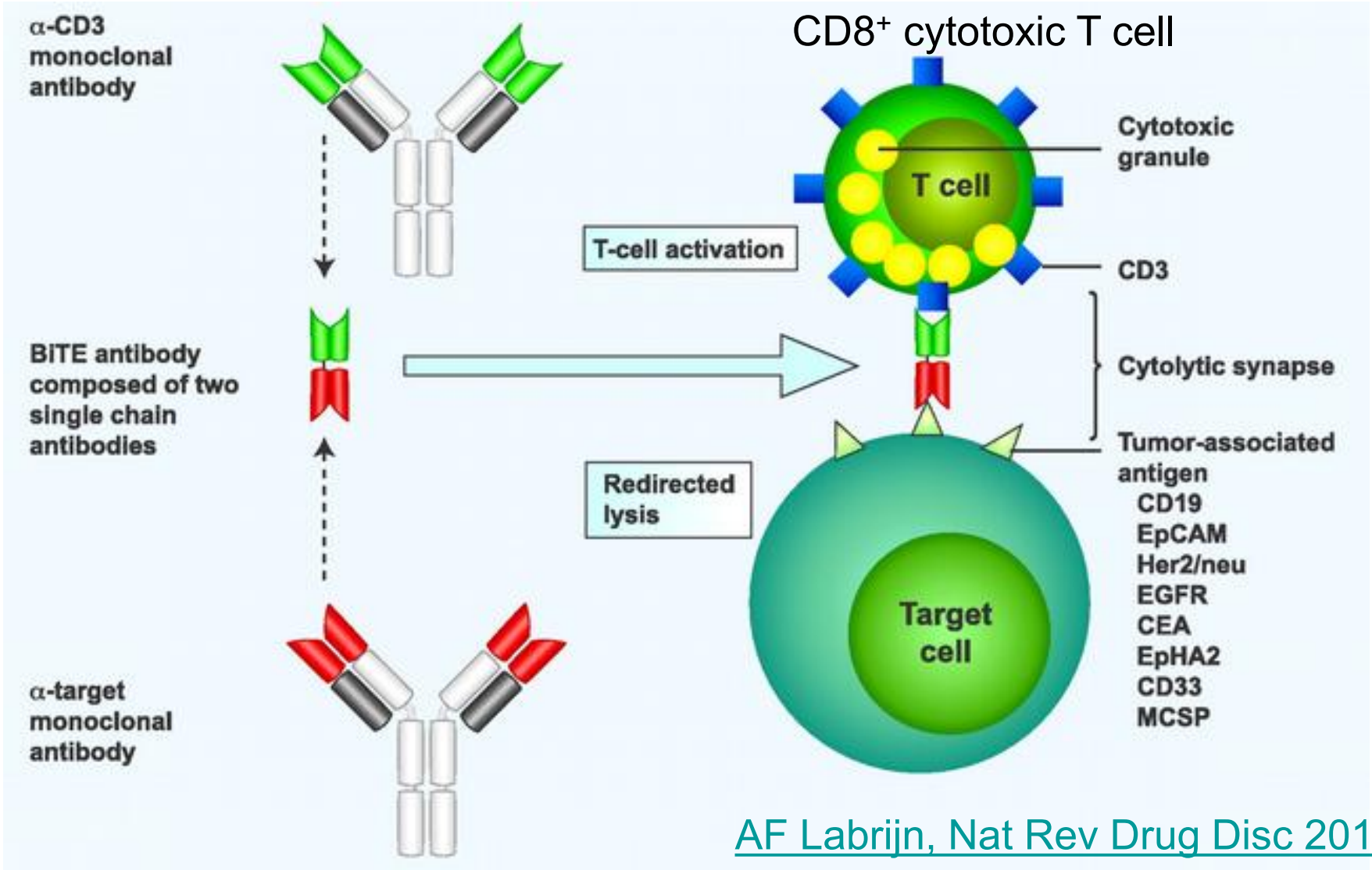
Compared to antibody therapy, small molecules are:
cheaper to manufacture (but may not be cheaper to the consumer)
simpler to administer (usually oral)
limited to surface-exposed targets

but
- short circulation
- less reliable pipeline
- lower success rate in trials



What to engineer: Fusions
(i) Proteins

Bispecific T cell engaging (BiTE) antibody successful as cancer therapeutic.



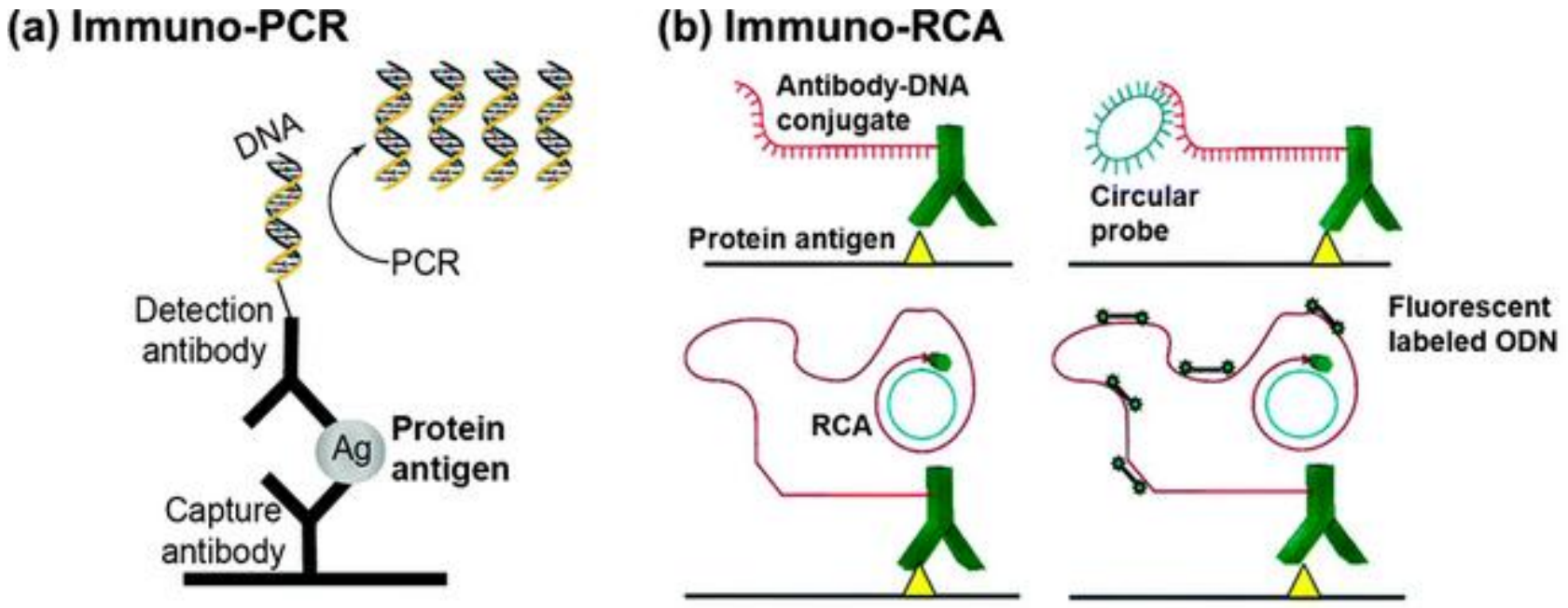
AF Labrijn, Nat Rev Drug Disc 2019

Why might it be useful to link DNA to an antibody?

A. Help signal amplification
B. Make more soluble
C. Tether to cell-surface
D. Assist in the target recognition

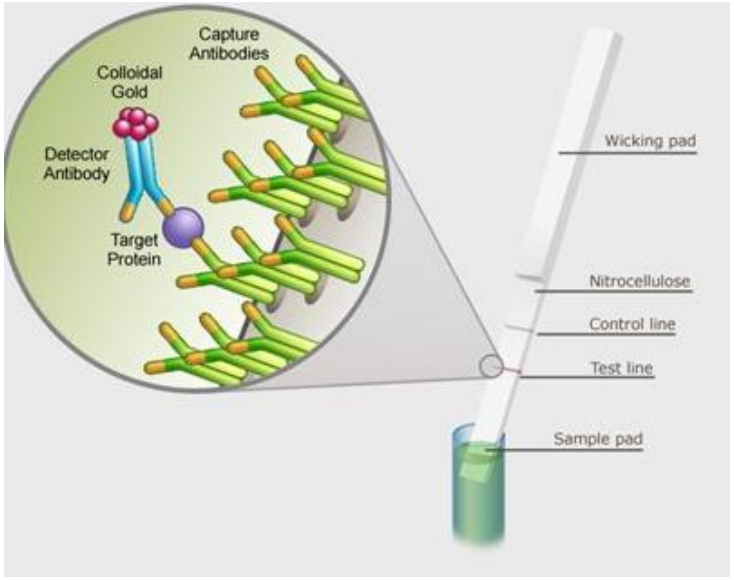
What to engineer: Fusions
(ii) Other biomolecules

Antibody-DNA conjugate for sensitive and multiplexed detection

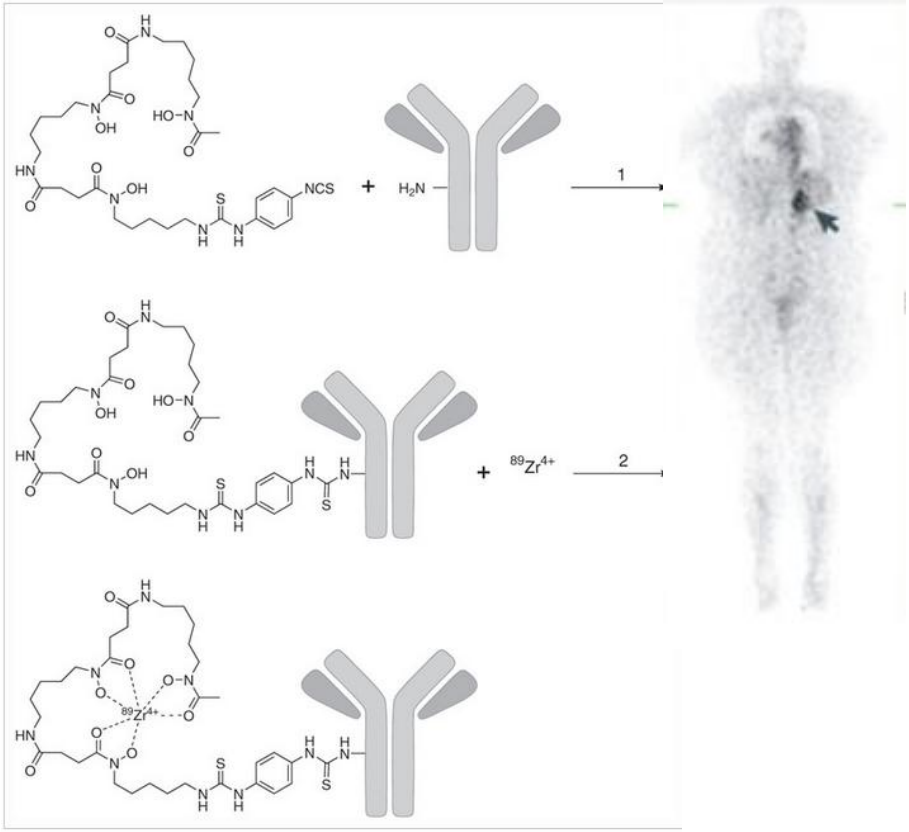


What to engineer: Fusions
(iii) Non-biological molecules

Large particle
for lateral flow assay

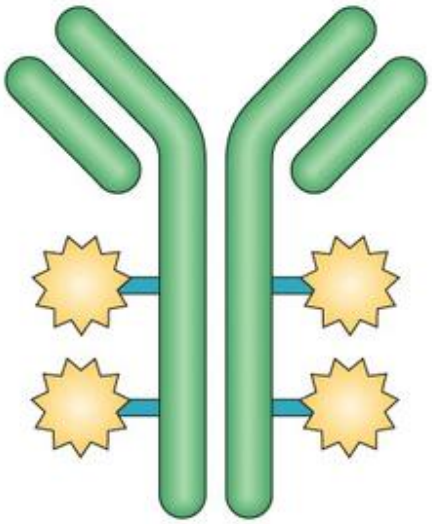


Small molecule for PET imaging



Bioconjugate Chemistry 2nd ed, G. Hermanson

Antibody drug-conjugate (ADC) design



Let's get the best of antibody specificity and small molecule potency!

ADCs have antibody-linker-drug
for effector function beyond natural antibodies
in killing cancer cells.

Extensive work optimising ADC-

- where to attach
- drug choice
- how many drug copies
- linker release

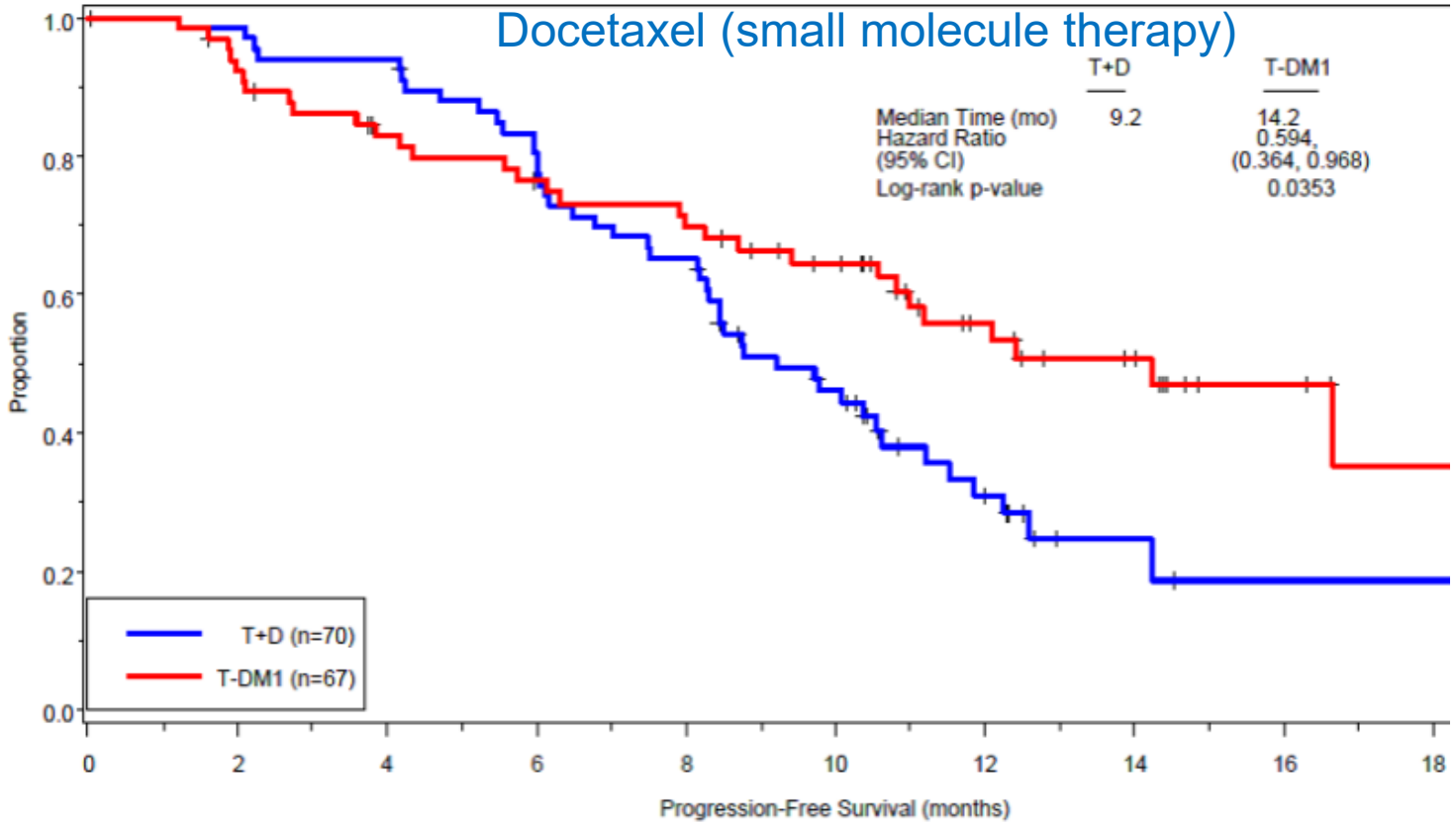
JZ Drago, Nat Rev Clin Onc 2021

What to engineer: Fusions
(iii) Non-biological molecules- toxins

HER2 is a protein on the surface of breast cancer that can be targeted by an antibody. Clinical trial result:

Anti-HER2 + Docetaxel (small molecule therapy)

Anti-HER2-drug

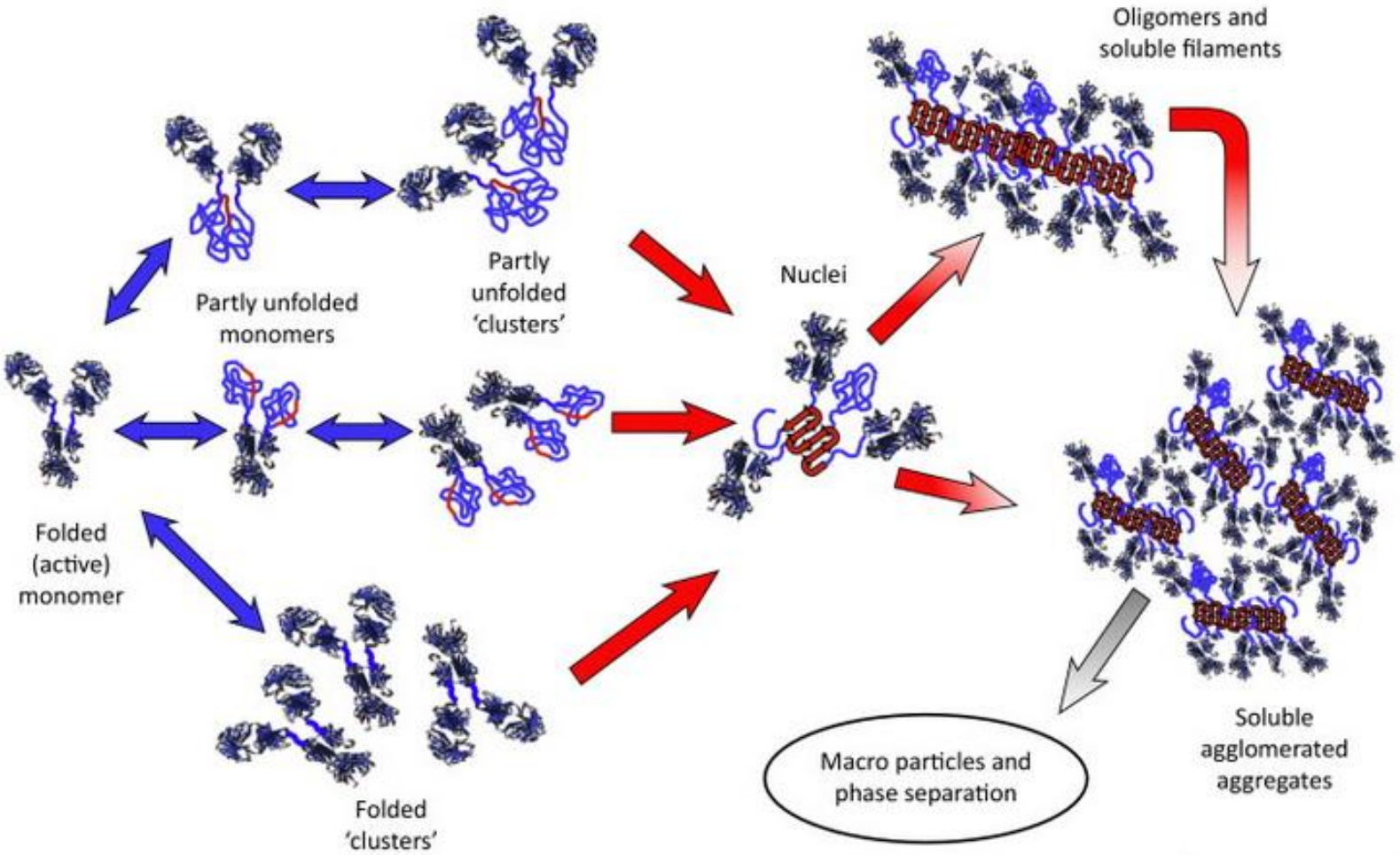


Treating a patient with a course of the anti-HER2 antibody-drug conjugate Kadcyla costs?

A. £90
B. £900
C. £9,000
D. £90,000



What to engineer: Multimerization

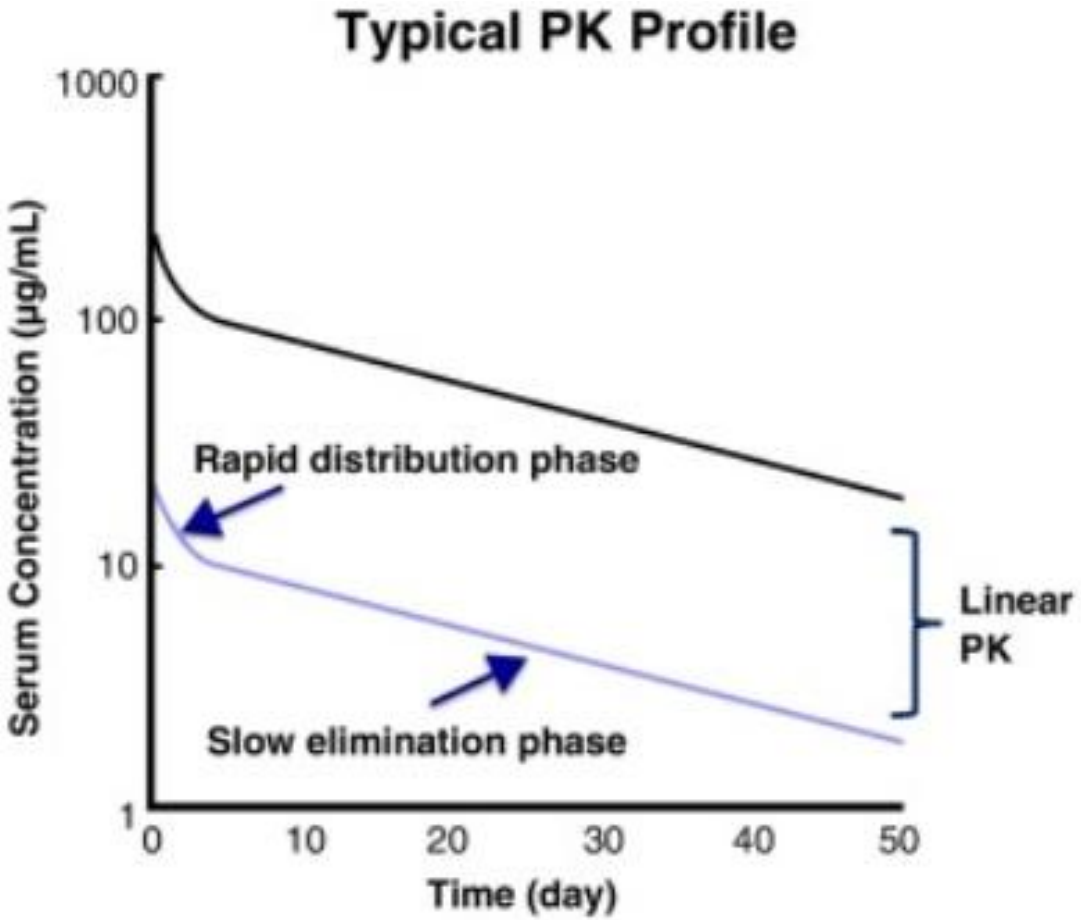


Everything sticks to everything else,
at high enough concentration...
eventually...

Antibody pharmacokinetics

After intravenous delivery
(note y axis is log)
Biphasic change:
1. Distribution
from blood to
tissue institial space
2. Elimination-
clearance by lysosomal deg.

No clearance by kidneys
(150 kDa but kidney ~20 kDa
limit)
Clearance changed by:
- Anti-drug antibodies
- Clearance of Ab while bound to target e.g. HER2

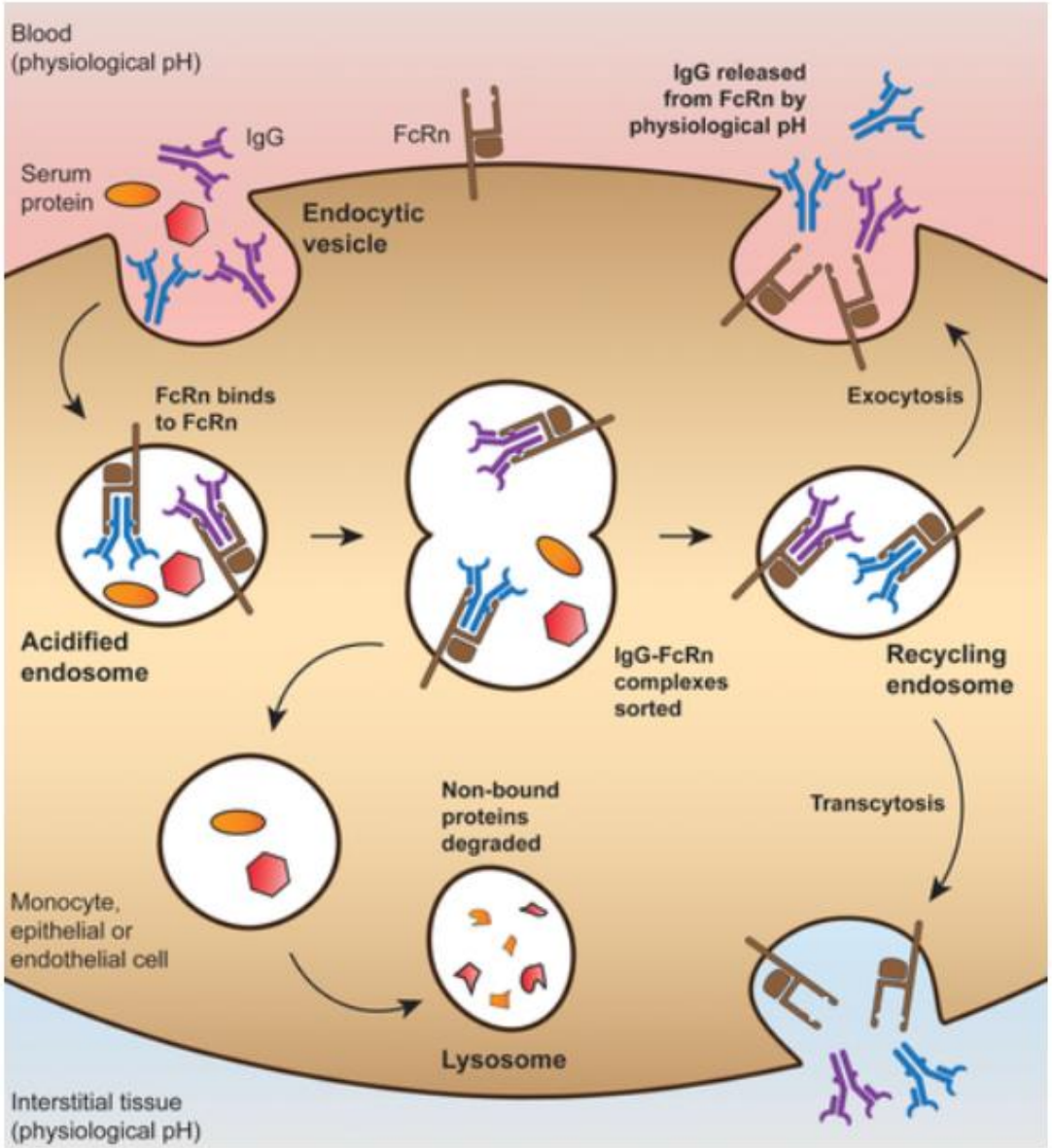


AV Kamath, Drug Disc Today 2015

Antibody pharmacokinetics- how antibody circulates for so long

FcRn binds IgG at endosomal pH, shuttling IgG back to surface, away from destruction in lysosomes (salvage pathway).

Any protein can be fused to Fc to increase half-life e.g. Etanercept TNFR extracellular domain fused to IgG Fc for rheumatoid arthritis therapy.



Antibody glycosylation

For a drug, want to know
- exactly what delivering
- delivering the same each time.

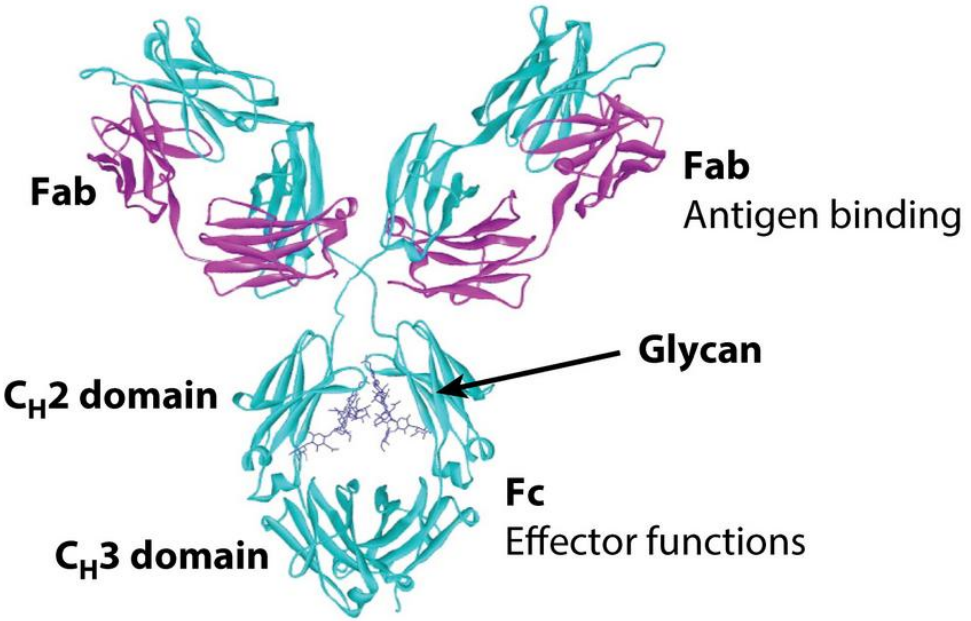
Difficult for protein therapeutics because glycosylation variable.

Depends on sequence,
cell-line and growth conditions.



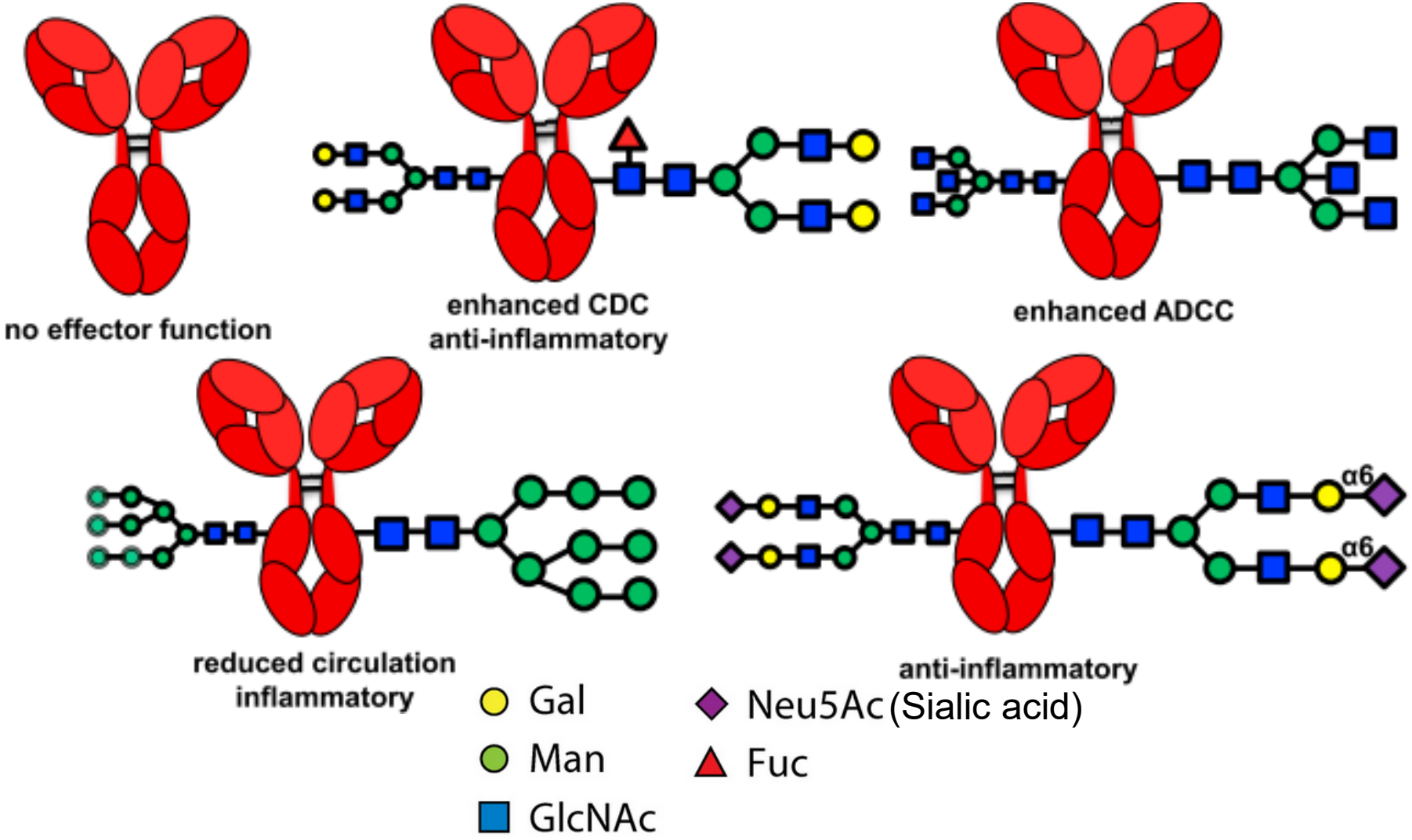
IgG has a glycan on
each C_H2 domain
with various sequences.
Study by mass spectrometry (MS).

LX Wang, Ann Rev Biochem 2019



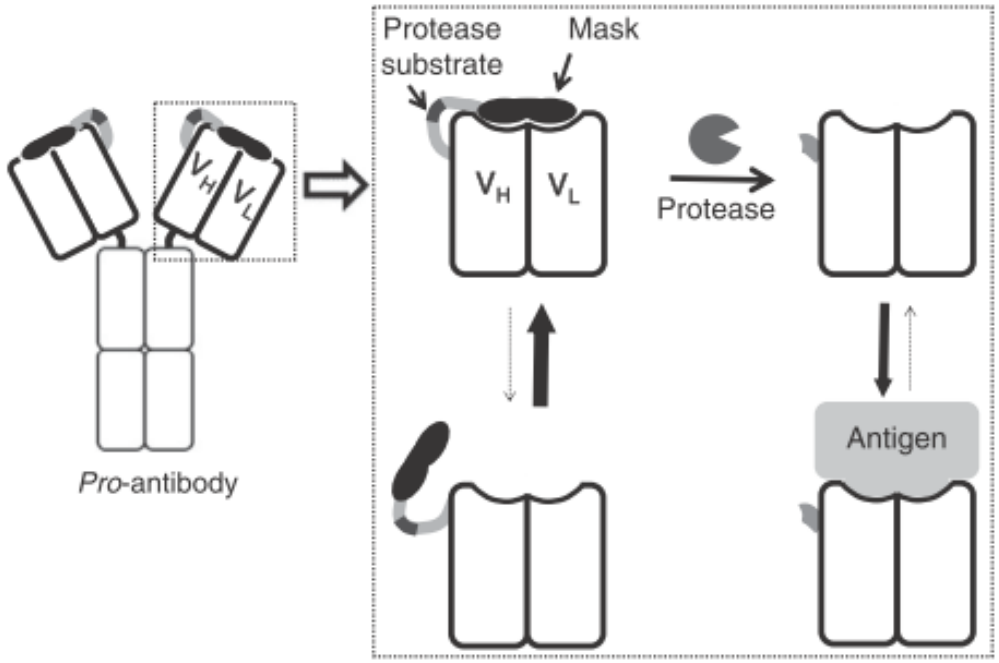
Glycans change antibody behaviour

Tuning glycosylation can transform antibody effects.



What to engineer: Switchable function

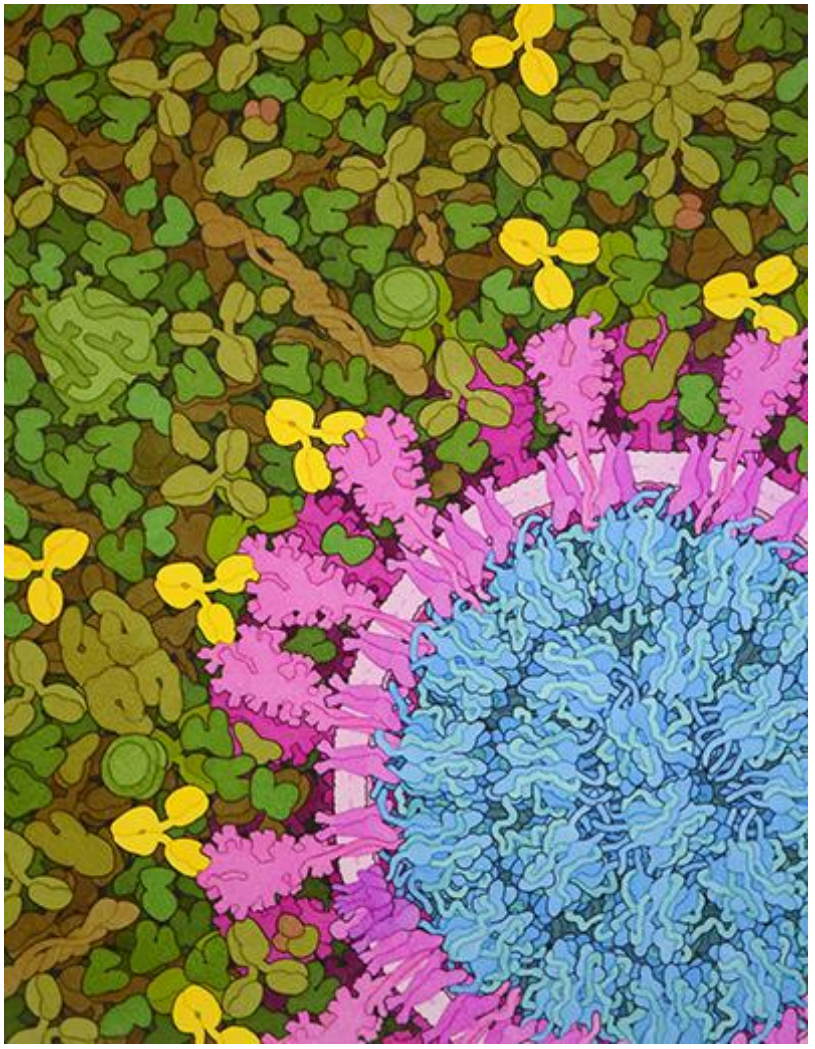
Important cancer targets can also be expressed on normal tissues, leading to dose-limiting toxicity.
How about a logic-gated Antibody which only binds if target present AND a cancer-associated protease present?



What to engineer: Activity -binding

Many ways to improve Antibody binding:
Dissociation constant of 1 μM OK,
best 100 pM

Need to be tight
AND
specific.



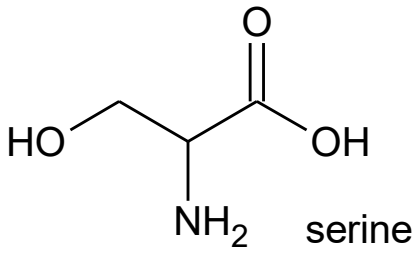
David Goodsell

Site-directed mutagenesis (SDM)

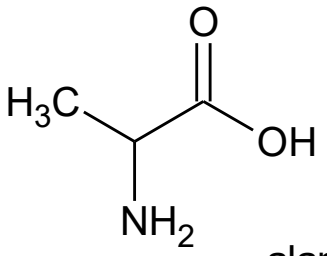


Replacement of one amino-acid by another, using genetic manipulation.

Goal: to explore function/ mechanism



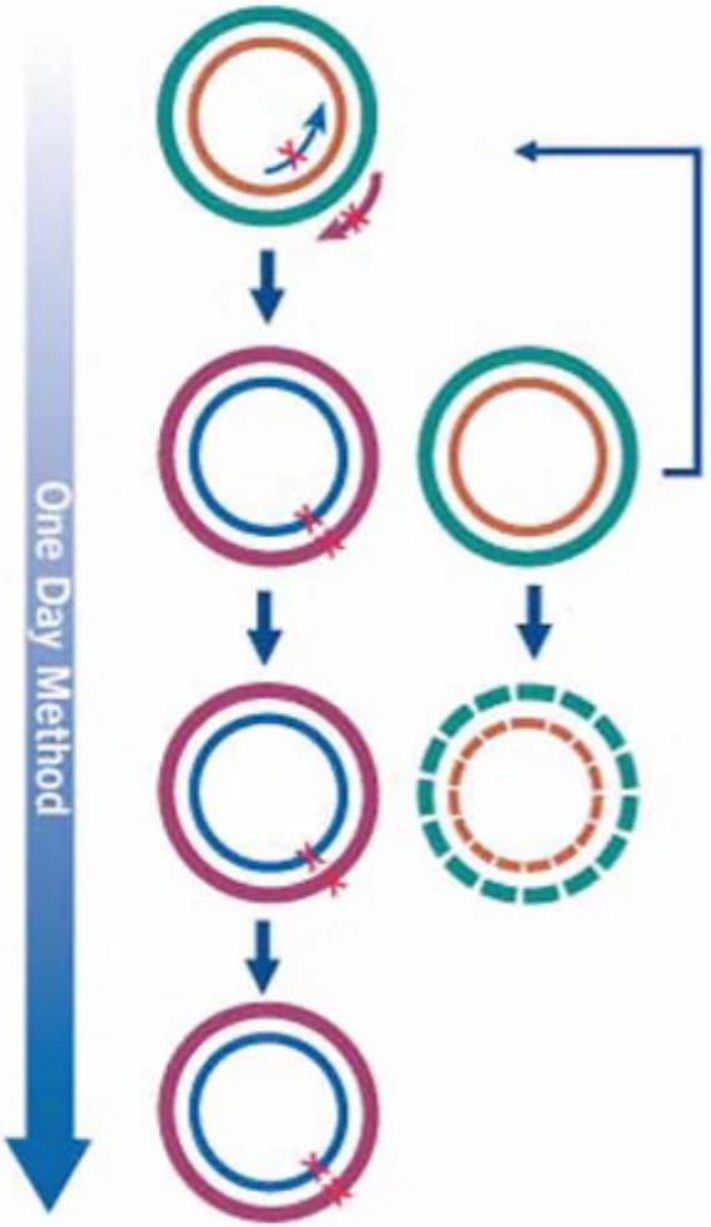
serine



alanine

e.g. removal of functional -CH_2OH and replacement by -CH_3.
Replacement of amino acid by a smaller one (no folding problems)

QuikChange SDM method



1. Mutant Strand Synthesis
Perform thermal cycling to:
- Denature DNA template
- Anneal mutagenic primers containing desired mutation
- Extend and incorporate primers with *high-fidelity* DNA polymerase

2. *Dpn*I Digestion of Template
Digest parental methylated and hemimethylated DNA with *Dpn* I

3. Transformation
Transform mutated molecule into competent cells for nick repair

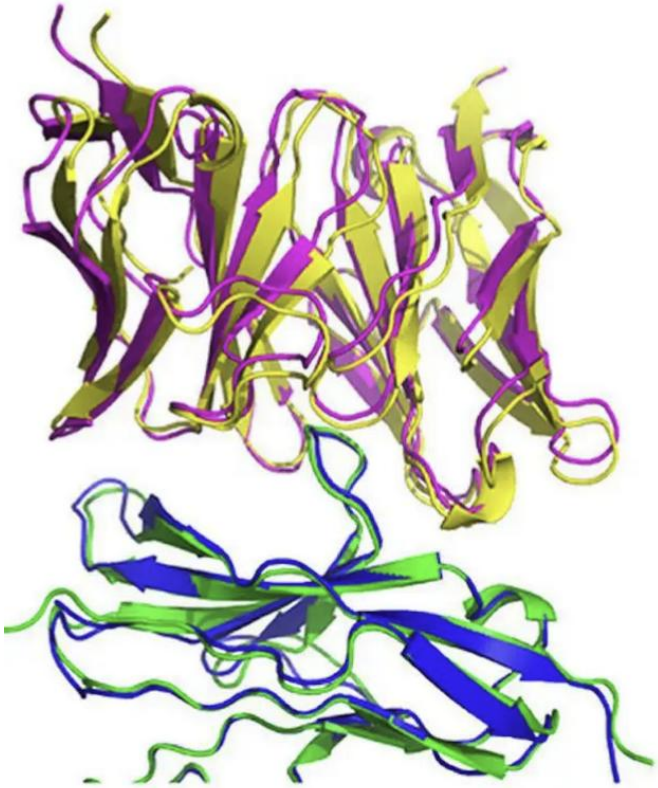
Note: superseded approach described in many textbooks.

Fast,
>90% success rate.
Single aa change, but also insert or delete a few residues.

Which amino acid to change?

1. Structure (e.g. with ligand) by X-ray crystallography or CryoEM
2. AlphaFold prediction of structure and complex

3. Conserved residues
4. Ala scanning
5. Stop codon to truncate or remove domain



Point mutants can be products

- Serine protease in biological laundry detergents

  Usually subtilisin, a non-specific protease

  With bleach, a key Methionine gets oxidised and enzyme inactive

  Methionine to Alanine mutant is resistant

(Important enzymes evolved by directed, combined with rounds of random evolution, e.g. for function at low temperature)



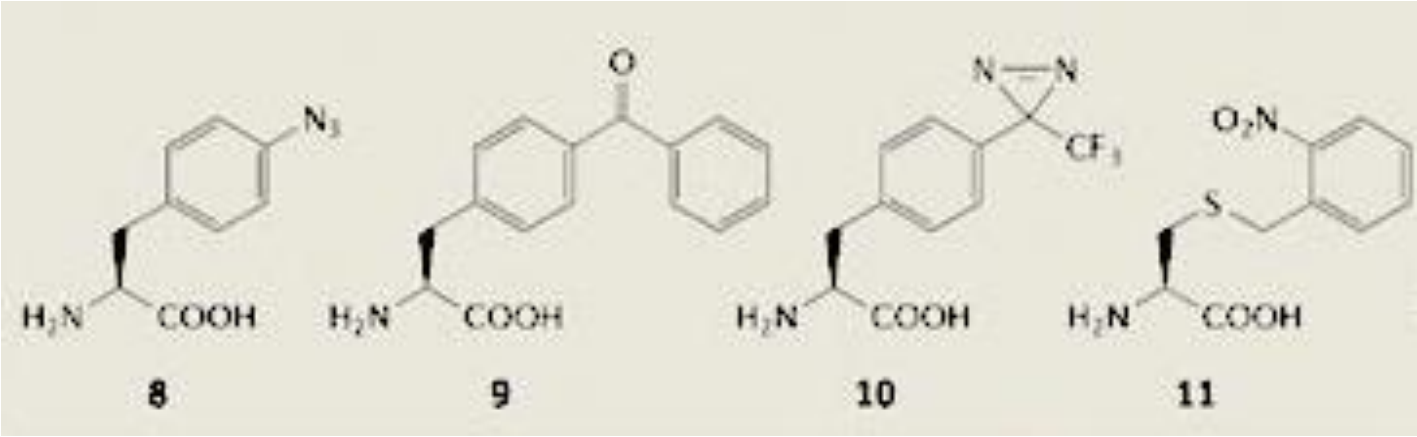
Modifying genetic code



Peter Schultz

Go beyond natural 20 amino acids
new chemical functionality
(e.g. photoswitching, photocaging, click chemistry)
subtle changes to side-chains

Examples of artificial amino acids incorporated (relatives of Tyrosine)



By *in vitro* translation
In *E. coli* (most common)
In yeast and mammalian systems more recently.
JW Chin Nat Rev Genetics 2021

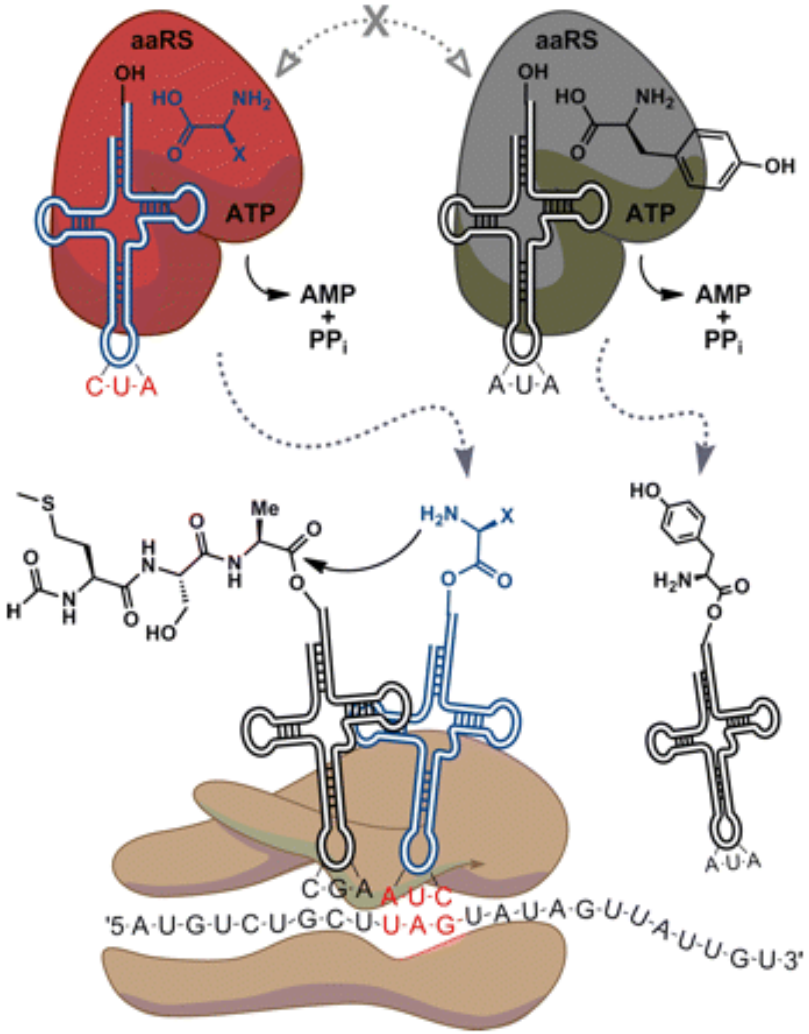
Artificial amino acid mutagenesis

In *E. coli*, get artificial tRNA to recognise stop codon UAG. Engineered aminoacyl tRNA synthetase recognises this tRNA and attaches artificial amino acid (and not any natural amino acid)

Engineered aaRS

Suppressor tRNA

Ribosome incorporates artificial aa into protein instead of terminating



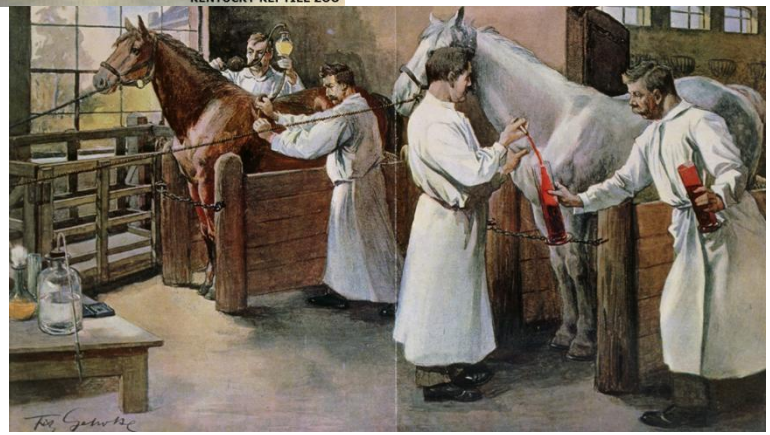
How do you generate an antibody?

Polyclonal antibodies:

1. Antiserum from animal

2. Human plasma





Monoclonal antibody (mAb):

1. Hybridoma

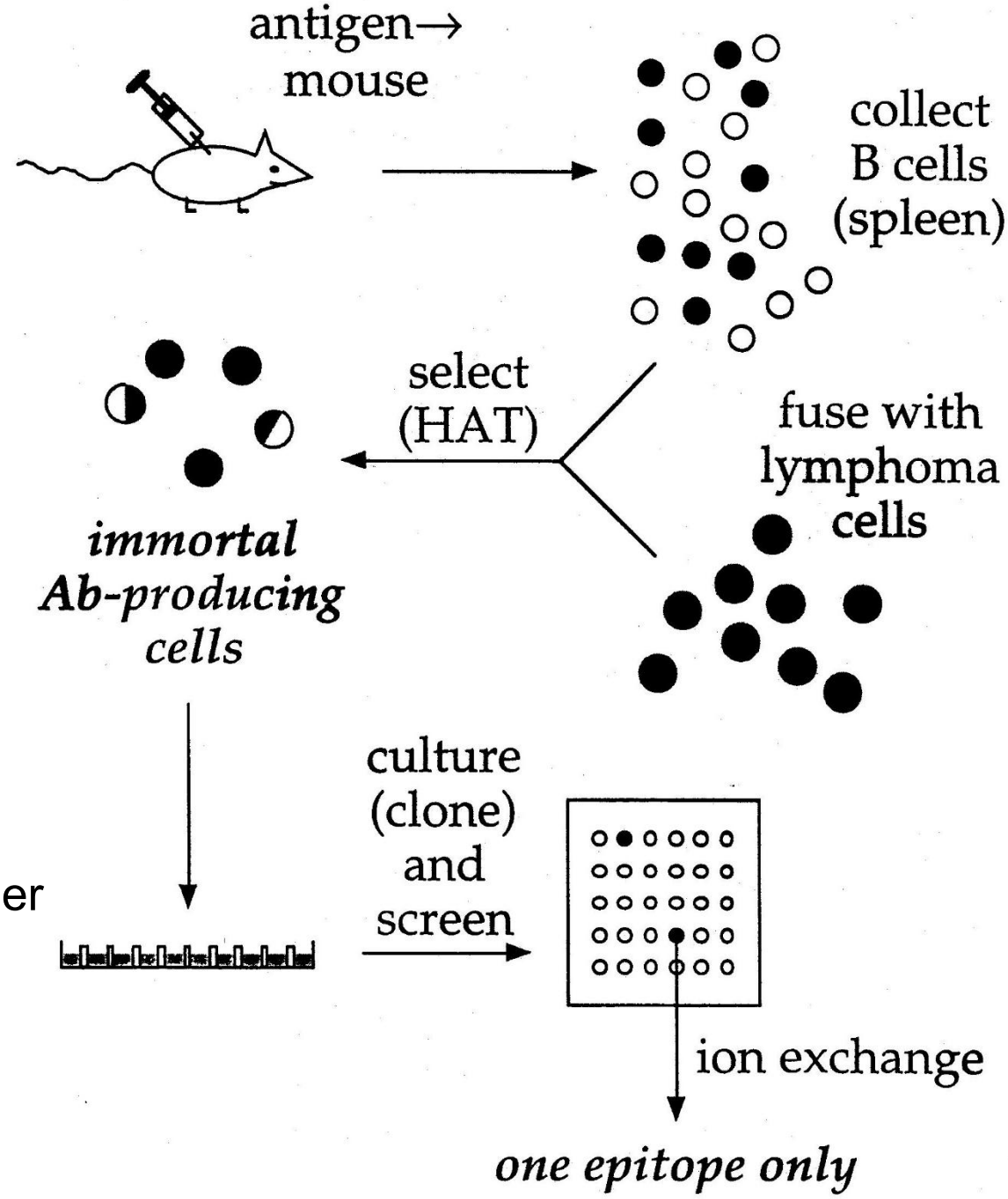
2. Phage

3. Humanized mice

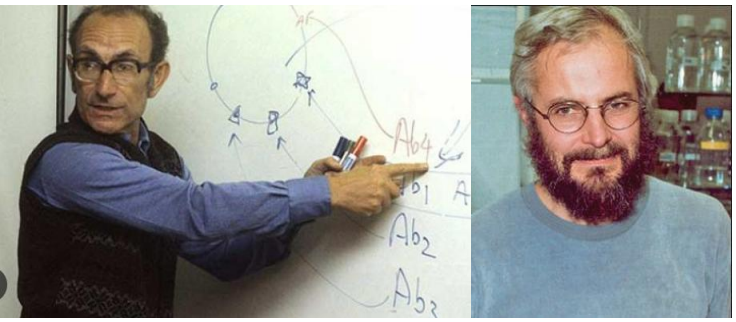
4. Direct from people

5. Computer prediction

MONOCLONAL ANTIBODIES

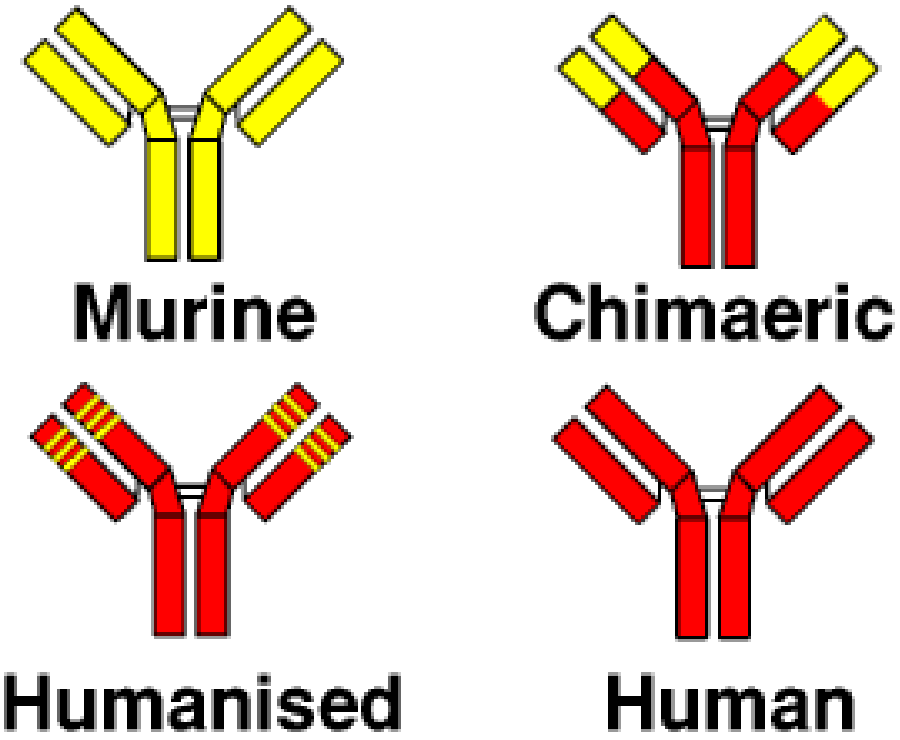


Cesar Milstein Georges Kohler



Antibody humanisation

Mouse mAbs not successful therapy in humans:
- anti-mouse immune response blocks repeated dosing
- effector function better with human Fc



Variable domains from mouse

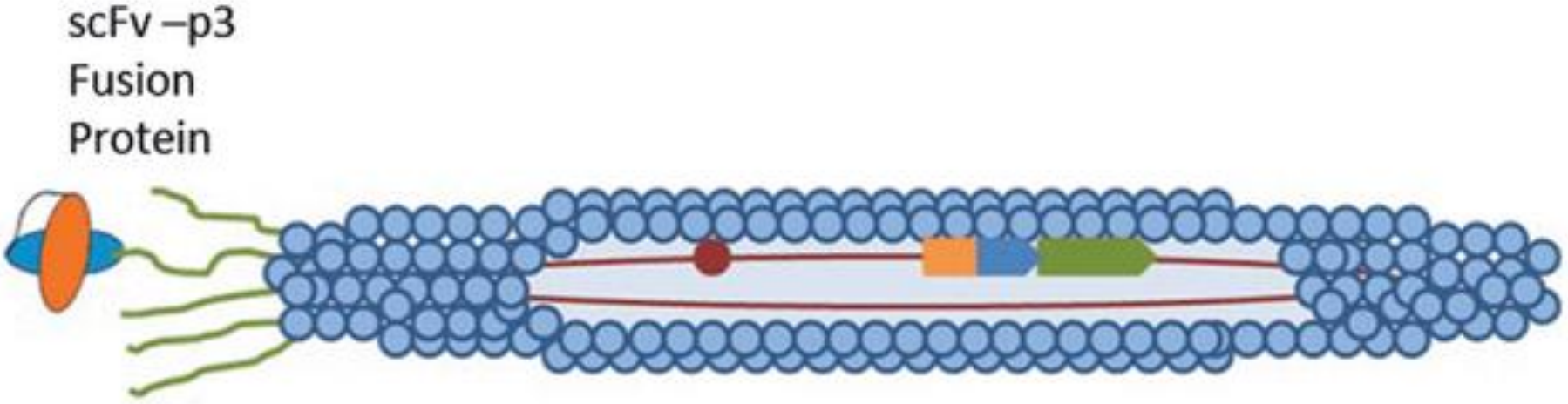
CDRs from mouse

Antibody generation by phage display

No need for animals!

speed, no worries about self-reaction

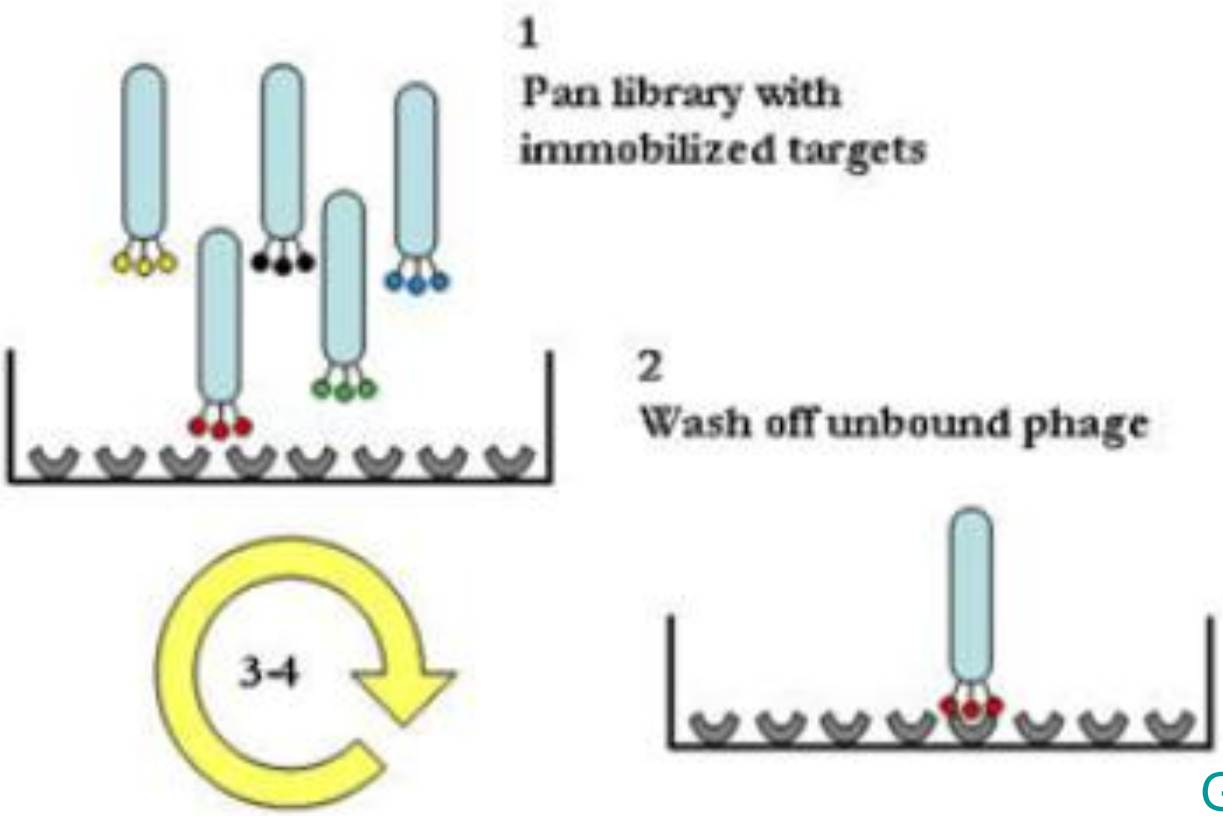
Make a library in a test-tube from HUMAN antibody sequences.
M13 is a filamentous phage that can display scFv antibody fragment on a coat protein.
Phenotype linked to genotype.



Bacteriophage with scFv displayed on p3

Antibody generation by phage display

1. Pan at least 1 billion library of human scFv phage against immobilised target.
2. Wash hard and recover good binders.
3. Phage are amplified in *E. coli*. Repeat cycle.
4. Sequence to find good scFv.
5. Reformat as IgG.



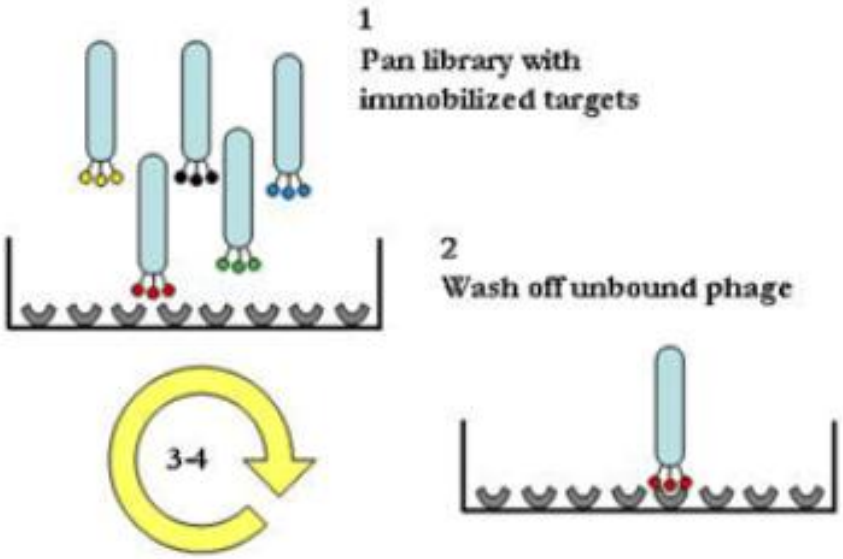
Sir Greg Winter,
Nobel Prize 2018



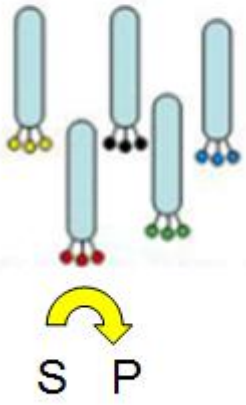
G. Winter, Ang Chem 2019

Challenge of linking genotype to phenotype

Easy to link gene to protein it expresses
(cellular expression, phage display, yeast display, ribosome display)



Easy to select for binding.
(Phage panning)

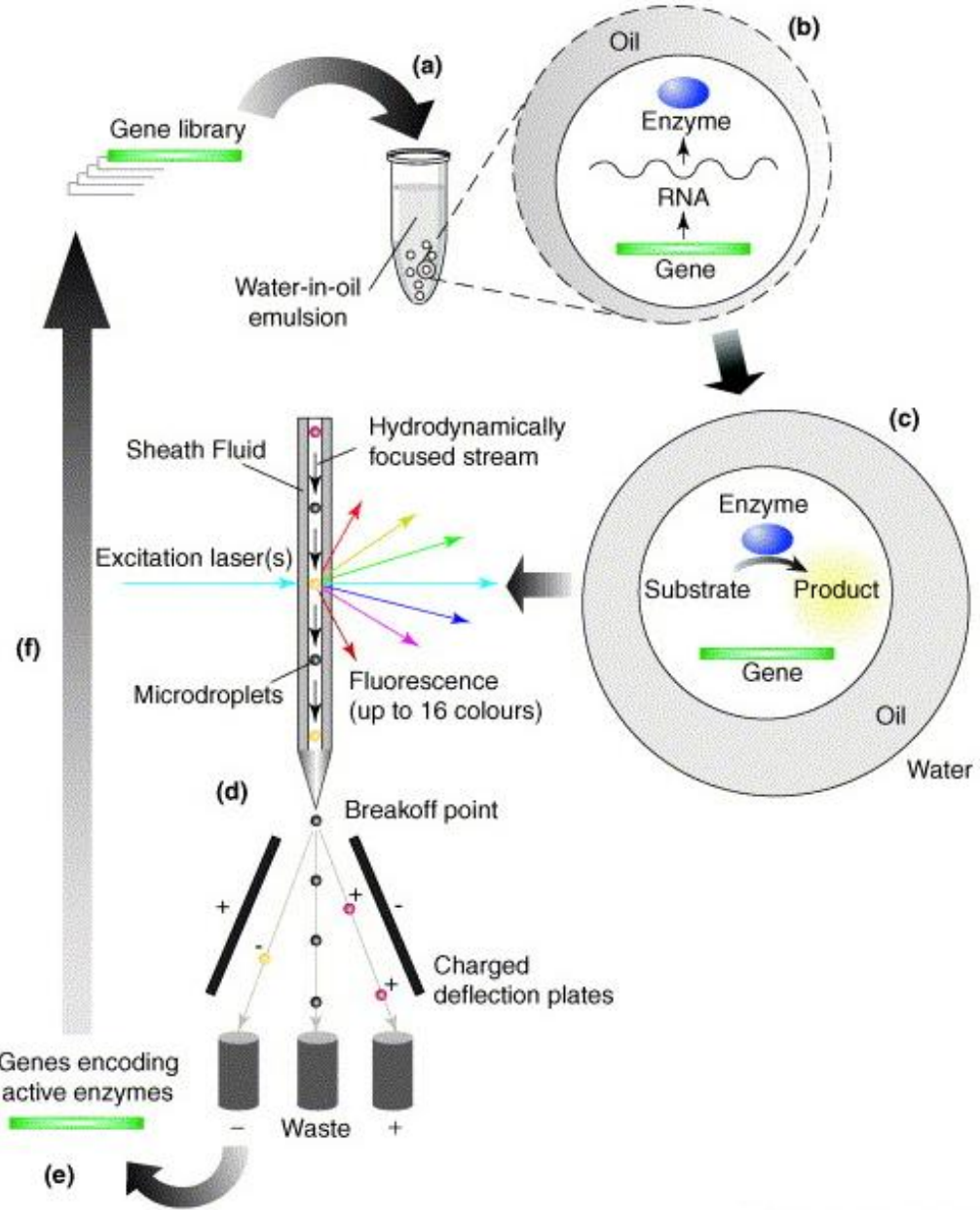


Hard to link gene
to enzymatic product

Size of protein space:

20 amino acids, 20^2 dipeptides... ~20^{200} decent-sized proteins
but only ~10^{80} atoms in the universe so need to search wisely!

In vitro compartmentalisation



(a) 0 or 1 gene per oil droplet

(b) in vitro transcription and translation within that droplet

(c) Enzymatic turn-over in droplet on fluorogenic substrate

(d) Flow cytometry (FACS) to isolate highly fluorescent droplets

(e) Large scale expression or re-mutation of active genes

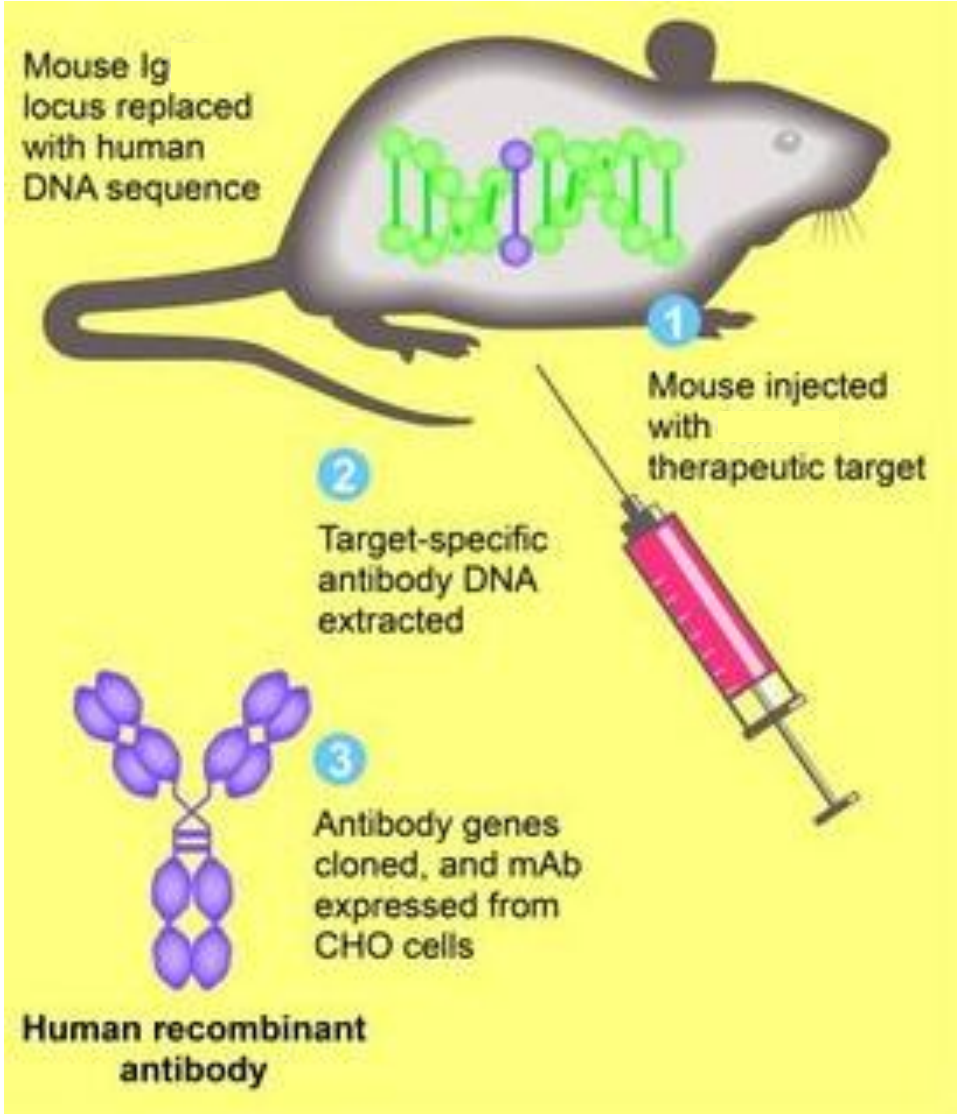
Gantz, Hollfelder 2024

Antibody generation by transgenic mice

GM mice with fully human antibody germline repertoire.

Conventional immunization and then affinity maturation by somatic hypermutation.

Sequence B cells to recover human antibody.



How much is a breeding mouse pair with a humanised antibody repertoire worth?

(A) $2,000

(B) $200,000

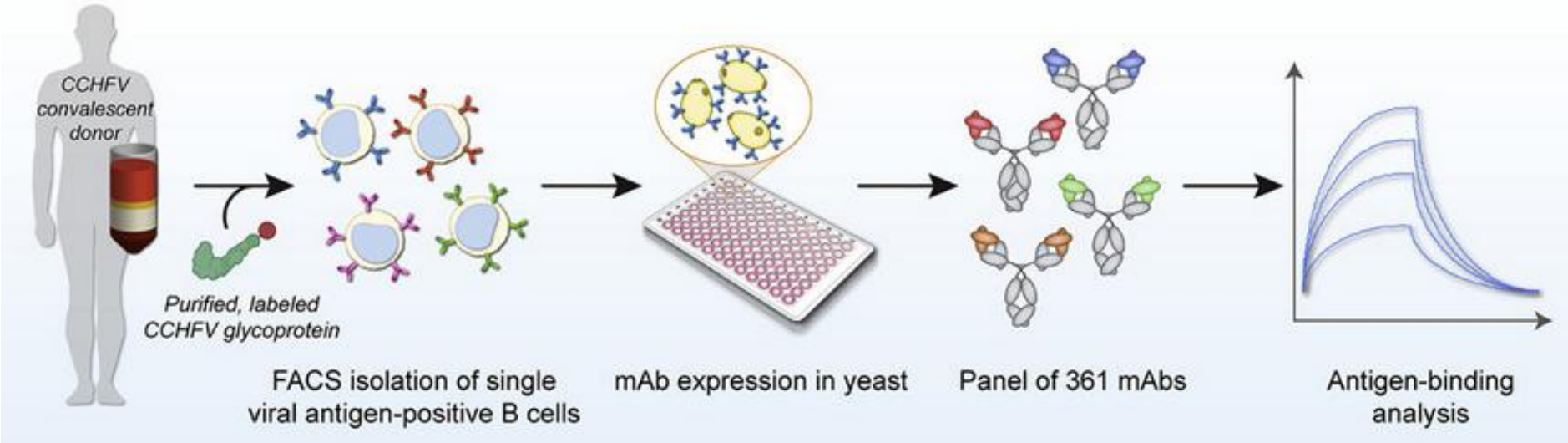
(C) $2,000,000

(D) $2,000,000,000

Antibody generation direct from people

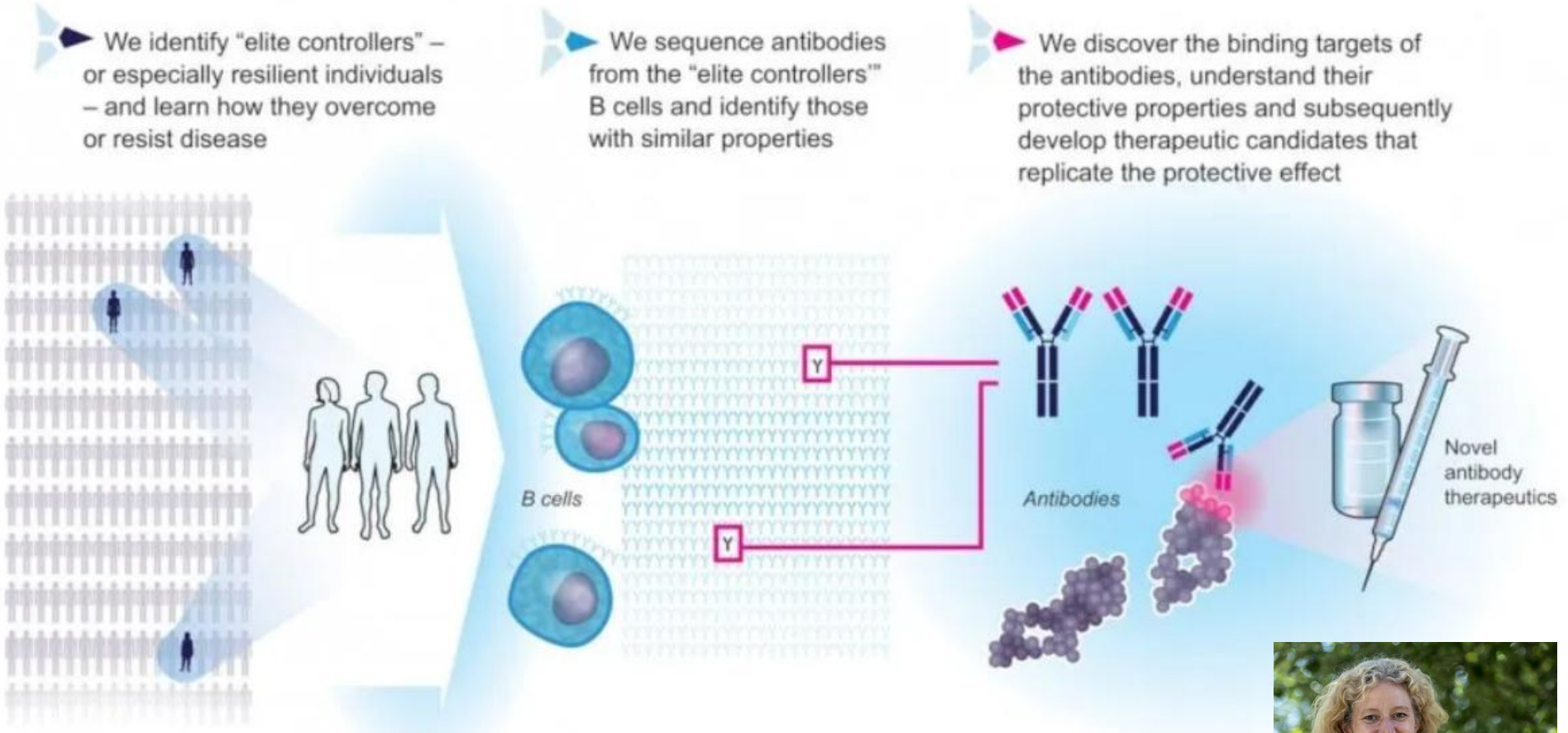
Can get human antibodies from humans!
Isolate relevant B cells and find relevant sequences.
Test recombinant Abs for
(i) binding
(ii) potency-
neutralisation





A Pedrioli, Trends Immunol 2021

Antibody generation from unusually healthy people! Alchemab



Unexpected targets (including self-proteins) or unexpected target sites.
Pipeline in cancer, Alzheimer's, Covid.
[M. Eisenstein, Nat Biotech 2022]



Jane Osbourn, CEO

Antibody design using computation

Generative AI
RFdiffusion: same principles as neural networks making realistic photos like DALL-E
- Trained on Ab complex structures in PDB.
- Easier to make interface with flat surface, rather than Ab loops.

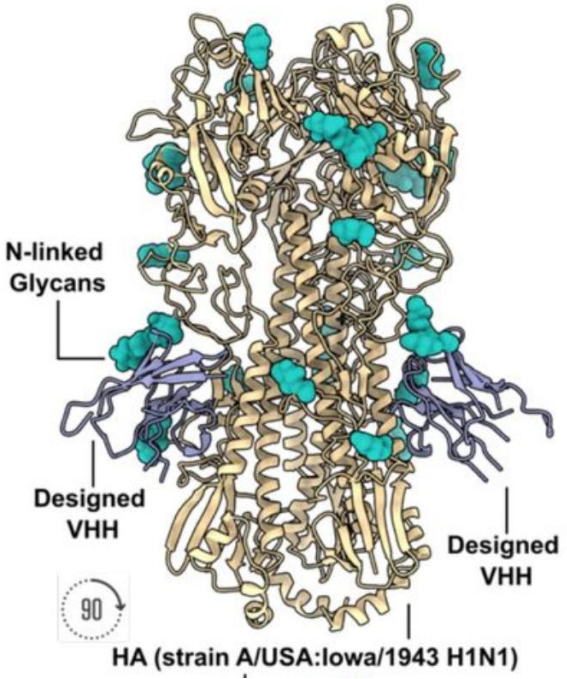


David Baker,
Nobel Prize 2024

Trained to Influenza target:
structure validated by CryoEM

Initially 1% antibodies bind, moderate affinity and not in clinical testing.
Ability changing fast!

NR Bennett, D Baker bioRxiv 2025



Antibody design AI competition



1. Affinity Maturation

design CDRs with improved affinity for SARS-CoV-2 RBD

2. Rank Prediction

identify sequences from Next Generation Sequencing (NGS) with highest affinity

3. NGS Inspired Design

with NGS dataset, design out-of-library Abs with highest affinity

MF Erasmus, Nature Biotech 2024

De novo design summary

Advantages
Start with no existing binder and no suitable bait
Applicable to challenging targets e.g. membrane-bound, toxic
"Library size" has no experimental limit but $>10^{18}$
Software publicly available

Disadvantages
Needs a lot of computer time-
Still may need wet-lab optimisation
Harder if substantial conformational changes

Computation is becoming an essential complement to all protein engineering.

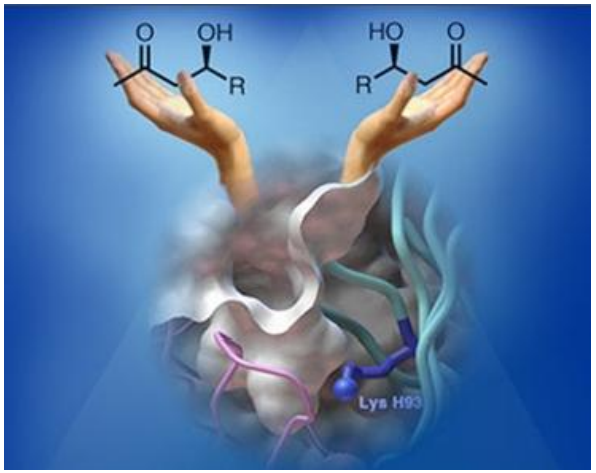
(NB: have to be accurate: 1 or 2 Å out can have big effect)

Summary

- Proteins can be chopped up or fused together
  made stable or labile
  respond to light or drugs or cellular signals

- Antibodies illustrate the great value of engineered proteins for diagnosis, therapy, and basic understanding of living systems
- Many remaining challenges in antibody engineering:
  better binding + logic + killing

- Protein engineering provides great opportunity for creativity in research
- There are many challenges for biology and society, which better engineered proteins could address.



Optional reading is listed on relevant slides with hyperlink.